PROJECT TITLE

VISITOR MANAGEMENT SYSTEM

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PROJECT REPORT

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Project Name	Visitor Management System
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Project Overview:

The Visitor Management System project in ServiceNow is developed to improve and automate the handling of network-related service requests within an organization. Businesses regularly receive requests for services such as network access, VPN activation, IP address allocation, visitor configuration, and troubleshooting connectivity issues. Managing these requests manually can lead to processing delays, inconsistent approvals, and limited visibility into the overall request workflow.

This solution utilizes the ServiceNow platform to automate the complete lifecycle of visitor requests. Employees can submit their requests through the Service Catalog, where automated workflows verify the submitted information, initiate approval processes, generate the required records, send email notifications, and continuously update the request status with minimal manual effort.

The project integrates various ServiceNow components, including Service Catalog, Catalog Variables, UI Policies, Custom Tables, Flow Designer, Approval Management, and Email Notifications, to build a reliable and scalable request management system. By automating these activities, the solution enhances operational efficiency, improves the user experience, ensures compliance with organizational standards, and maintains a complete audit trail for every visitor request.

Table of Contents:

1. Introduction :

In today's organizations, a stable and secure network infrastructure is essential for supporting day-to-day business activities. Employees frequently raise requests for services such as network access, VPN connectivity, IP address assignment, router and switch configuration, network relocation, and onboarding of new visitors. Handling these requests manually often requires considerable administrative effort and may result in delays, human errors, and inconsistent processing.

As businesses expand, the number of visitor requests continues to grow, making manual management increasingly inefficient. This can affect the timely resolution of requests, make it difficult to meet Service Level Agreements (SLAs), reduce visibility into request progress, and create inconsistencies in the approval process.

The **Visitor Management System** project, developed using the ServiceNow platform, provides a centralized and automated solution for managing visitor requests. Users can easily submit requests through the **Service Catalog**, while automated workflows validate the submitted information, route requests for approval, create the necessary records, send notification emails, and track the request status throughout its lifecycle.

The proposed system offers the following benefits:

- Faster processing of visitor requests
- Standardized and consistent workflow execution
- Reduced manual intervention and administrative workload
- Enhanced transparency throughout the request lifecycle
- Improved compliance with organizational policies
- Accurate monitoring and tracking of requests
- Increased user satisfaction through efficient service delivery

By utilizing the low-code capabilities of ServiceNow, this project demonstrates how **IT Service Management (ITSM)** can automate routine visitor operations while providing flexibility for future enhancements and organizational growth.

2. Project Objectives :

The main objective of this project is to develop an automated **Visitor Request Management System** using the ServiceNow platform that reduces manual effort and improves the efficiency of handling network-related service requests. The system is designed to provide users with a simple and centralized platform for submitting requests, while automating the processes of validation, approval, record generation, notifications, and request tracking.

The project also focuses on building a standardized workflow that enhances service quality, increases operational efficiency, and ensures timely request fulfillment with minimal human intervention.

The specific objectives of this project are:

- Develop a centralized **Service Catalog** for submitting various visitor requests.
- Design interactive request forms using **Catalog Variables** to capture the required information.
- Implement **UI Policies** to dynamically display or hide fields based on user input.
- Create **Custom Tables** to securely store and manage visitor request data.

- Configure **Flow Designer** to automate the complete request processing workflow.
- Establish automated approval mechanisms involving managers or network administrators.
- Generate email notifications automatically at different stages of the request lifecycle.
- Maintain detailed audit records for monitoring and tracking all submitted requests.
- Reduce request processing time while improving compliance with Service Level Agreements (SLAs).
- Build a scalable and flexible solution that can be extended to support additional visitors in the future.

3. Key Features:

The proposed system includes the following major features:

- User-friendly Service Catalog for submitting visitor requests.
- Dynamic forms with intelligent field visibility.
- Catalog Variables for capturing request information.
- Variable Sets for reusable user information.
- Automated approval workflows using Flow Designer.
- Custom Visitor Request Database table for request storage.
- Automatic creation of backend records.
- Email notifications during approval and completion stages.
- Real-time request status tracking.
- Audit-ready approval history.
- Scalable workflow architecture.
- Reduced manual processing time.

4. Project Milestones:

The project implementation follows five major milestones.

Milestone 1: Custom Table and Form Creation
Objective:
Create the database structure and visitor registration form for managing visitor information.
Activities:

Create the Visitor Management custom table.
Add required fields for visitor information.
Configure field properties and data types.
Create the Visitor Registration form.
Organize the form layout for easy data entry.

Milestone 2: Form Validation

Objective:

Ensure accurate and valid visitor information is entered into the system.

Activities:

Create a Client Script for phone number validation.
Configure mandatory fields.
Validate Indian mobile numbers.
Display validation messages for invalid input.
Prevent submission of incorrect data.

Milestone 3: Notification and Check-In Automation

Objective:

Automate visitor notifications and simplify the visitor check-in process.

Activities:

Create an Email Notification for the Security team.
Configure notification triggers.
Design the email template.
Create a Check In UI Action.
Update visitor status from Registered to Checked In automatically.

Milestone 4: Testing

Objective:

Verify that all features of the Visitor Management System function correctly.

Activities:

Test visitor registration.
Test phone number validation.
Test email notification delivery.
Test the Check In UI Action.
Verify visitor status updates.
Perform end-to-end system testing.

Milestone 5: Deployment

Objective:

Deploy the Visitor Management System for organizational use.

Activities:

Verify all configurations.
Publish the application.
Monitor system performance.
Train end users.
Ensure smooth operation after deployment.
This version is customized for your Visitor Management System and matches the style of the template you shared.

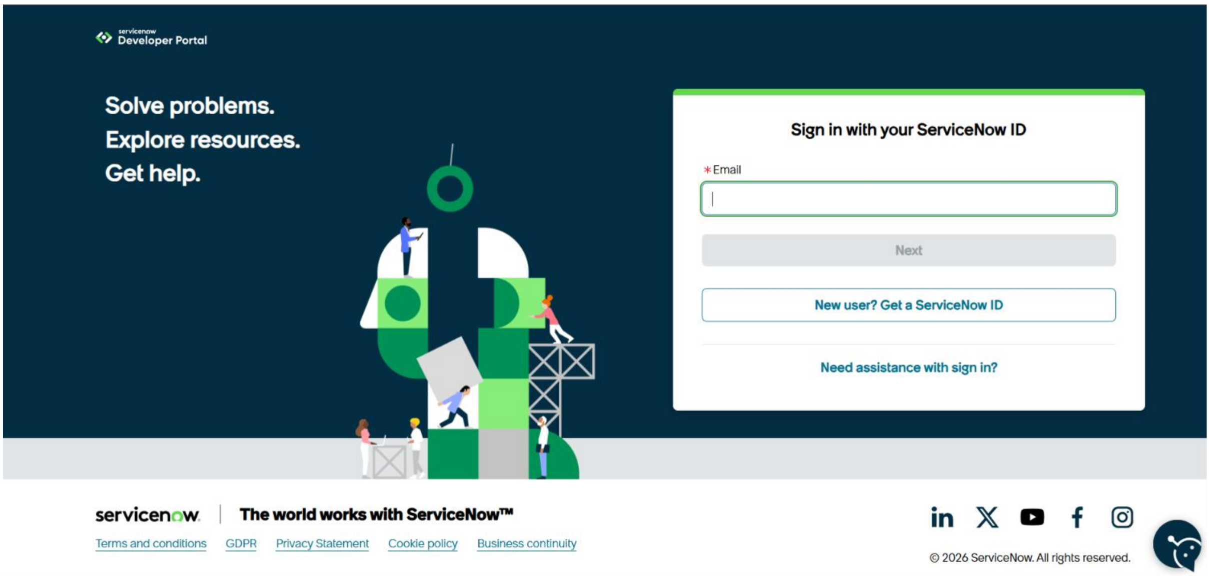
5. ServiceNow Developer Setup

Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

5.1 Creating a ServiceNow Developer Account :

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
    - First Name
    - Last Name
    - Email Address
    - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.



5.2 Accessing Creator Studio

Once the Personal Developer Instance is ready, ServiceNow redirects the developer to Creator Studio.

Creator Studio provides a no-code and low-code environment for building applications. It simplifies application development by allowing developers to create forms, workflows, tables, automation, and integrations through an intuitive graphical interface.

Creator Studio offers features such as:

--Creator Studio offers features such as:

--Application Creation

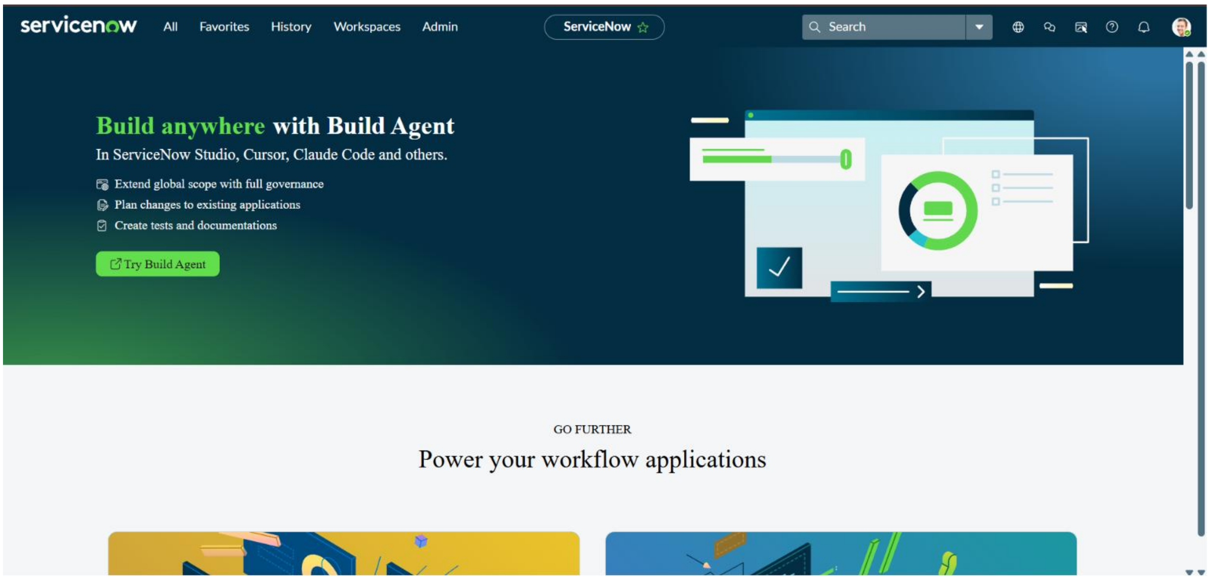
--Custom Table Creation

--Form Design

--Data Modeling

--Flow Automation

--Email Notification Configuration



6.1 Custom Table Creation:

Purpose:

The Visitor Management table is the core component of the Visitor Management System. It is created to store and manage all visitor-related information in a centralized location. Each record in this table represents a visitor request and contains details such as the visitor's name, contact information, company, visit date, host employee, purpose of visit, and current visit status. This table serves as the foundation for implementing validations, email notifications, and check-in functionality.

Navigation:

Application Navigator → System Definition → Tables → New

Steps to Create the Custom Table

1.Log in to your ServiceNow Developer Instance.

2.Open the Application Navigator.

3.Search for Tables.

4.Navigate to System Definition → Tables.

5.Click the New button.

6.Enter the following details:

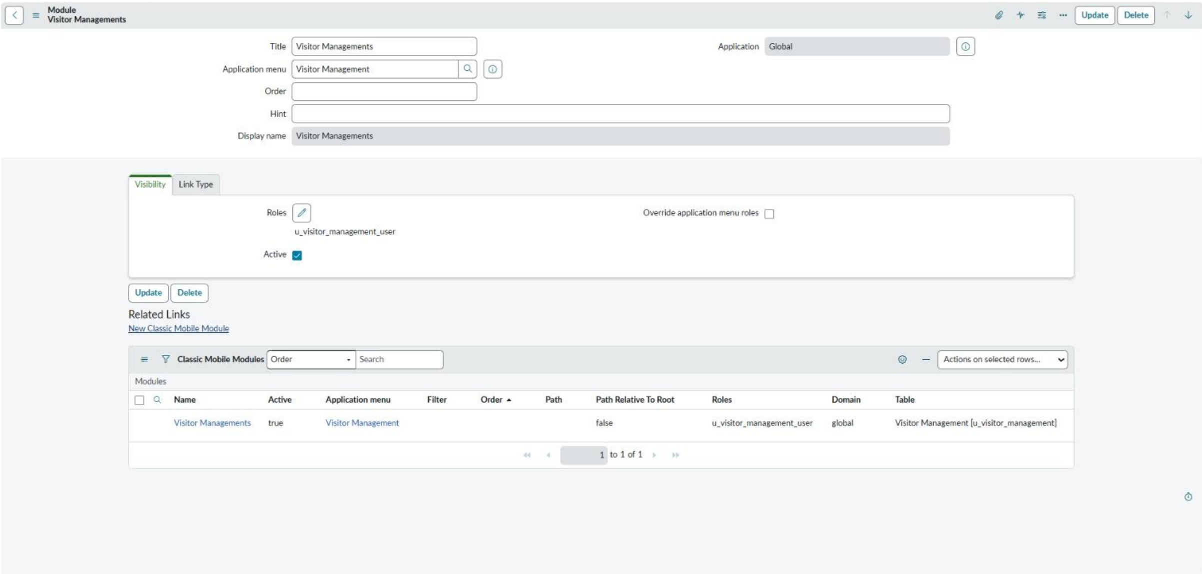
--**Field-value**- Visitor Management

--**Field-Name**- Visitor Management (Auto-generated)

--Auto Create Module-Yes

--Extends Table-None

7.Click the Submit button to create the custom table successfully.



6.2 Field Creation:

Purpose

After creating the Visitor Management table, the required fields are added to store complete visitor information. These fields capture essential details such as the visitor's name, phone number, email address, company, visit date, host employee, purpose of the visit, and current status. These fields enable efficient visitor registration, tracking, and management within the application.

Navigation

Application Navigator → System Definition → Tables → Visitor Management → Columns (Fields)

(Alternatively: Open the Visitor Management table and click Columns or New to add fields.)

Steps to Create the Fields

1.Open the Visitor Management table.

2.Navigate to the Columns section.

3.Click the New button to create a field.

4.Create the following fields one by one:

Field Label------Field Type

Visitor Name-------String

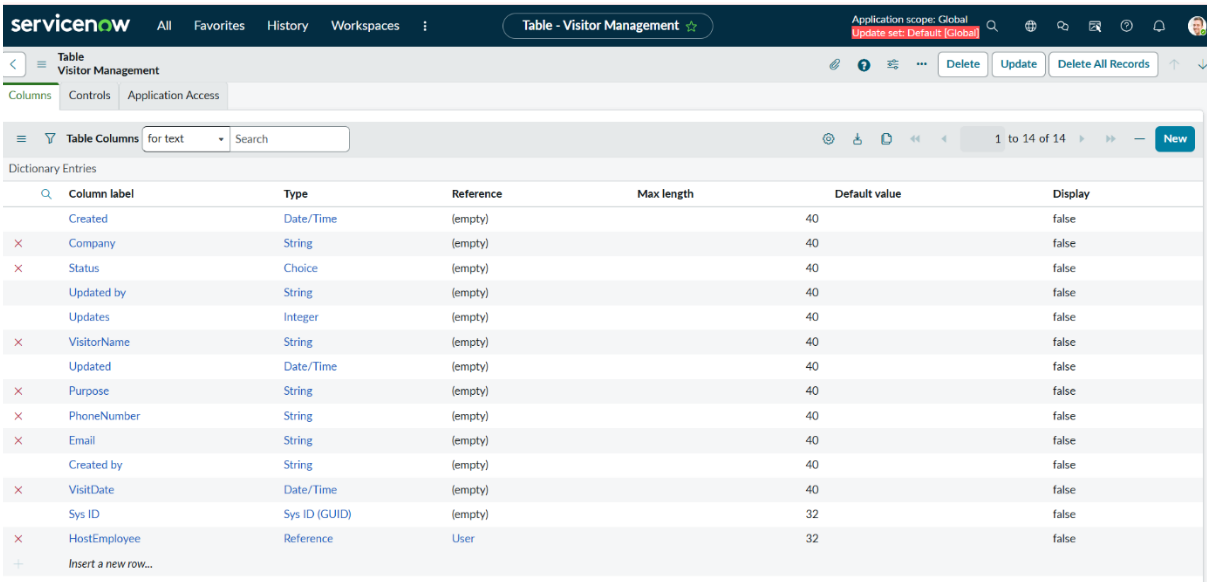
Phone Numbeer------String

Email------Email

Company------String

Visit Date-------Date/Time

Host Employee------Reference → User (sys_user)
Purpose-------String
Status--------Choice
For the Status field, add the following choice values:
Registered
Checked In
Checked Out
Click Submit or Save after creating each field.



6.3 Application Menu and Module Creation

Purpose:

The Application Menu provides a centralized location for accessing all modules related to the Visitor Management System. Creating the application menu and modules makes it easy for users to register new visitors and view all visitor records from the Application Navigator.

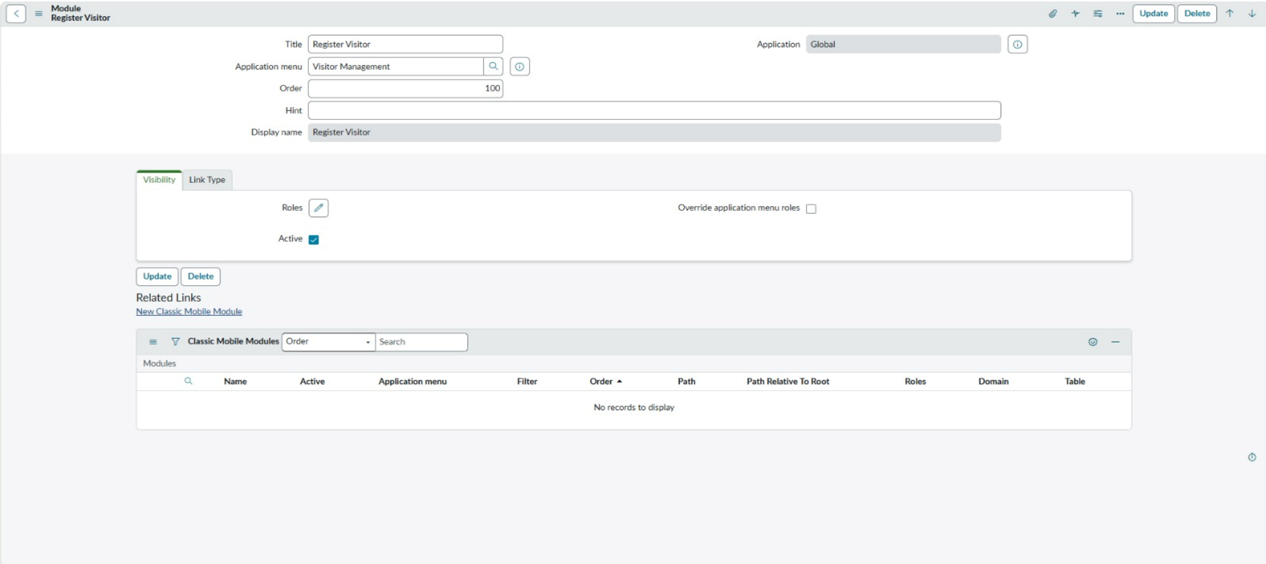
Navigation:

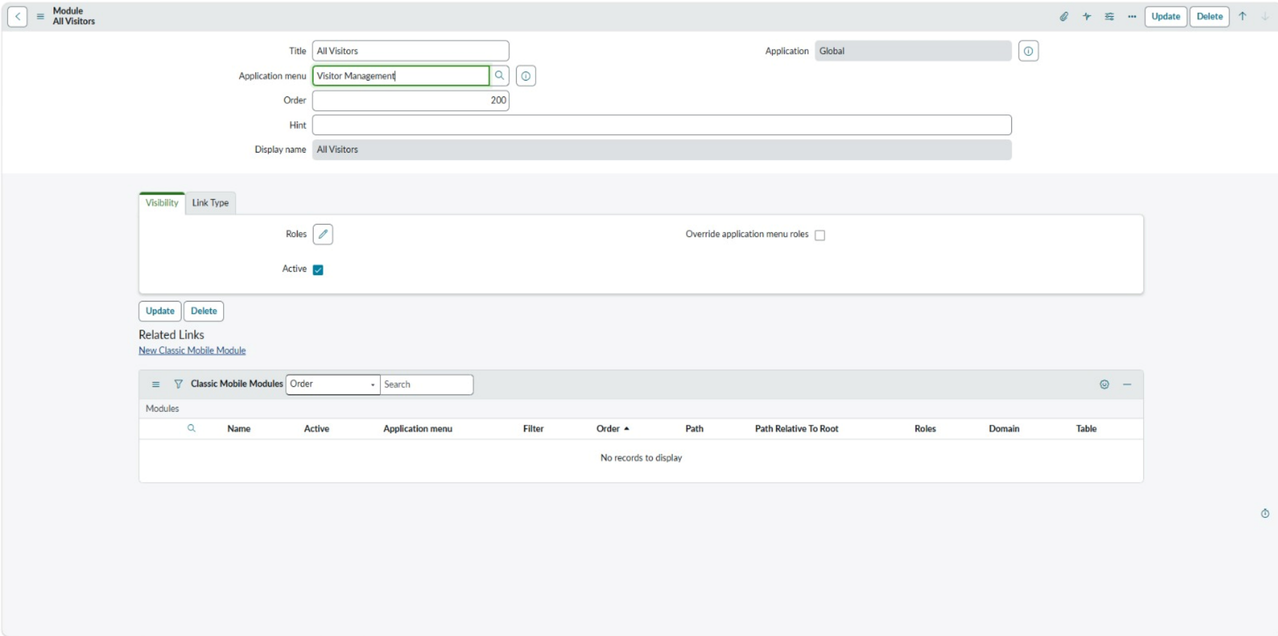
Application Navigator → Studio → Open Your Application

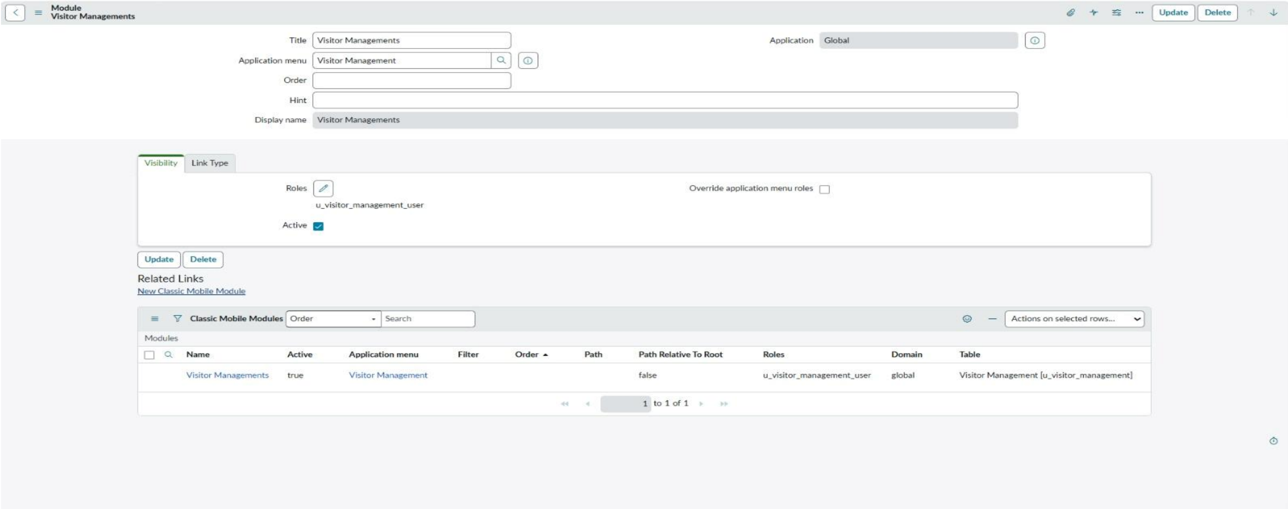
Steps to Create the Application Menu and Modules:

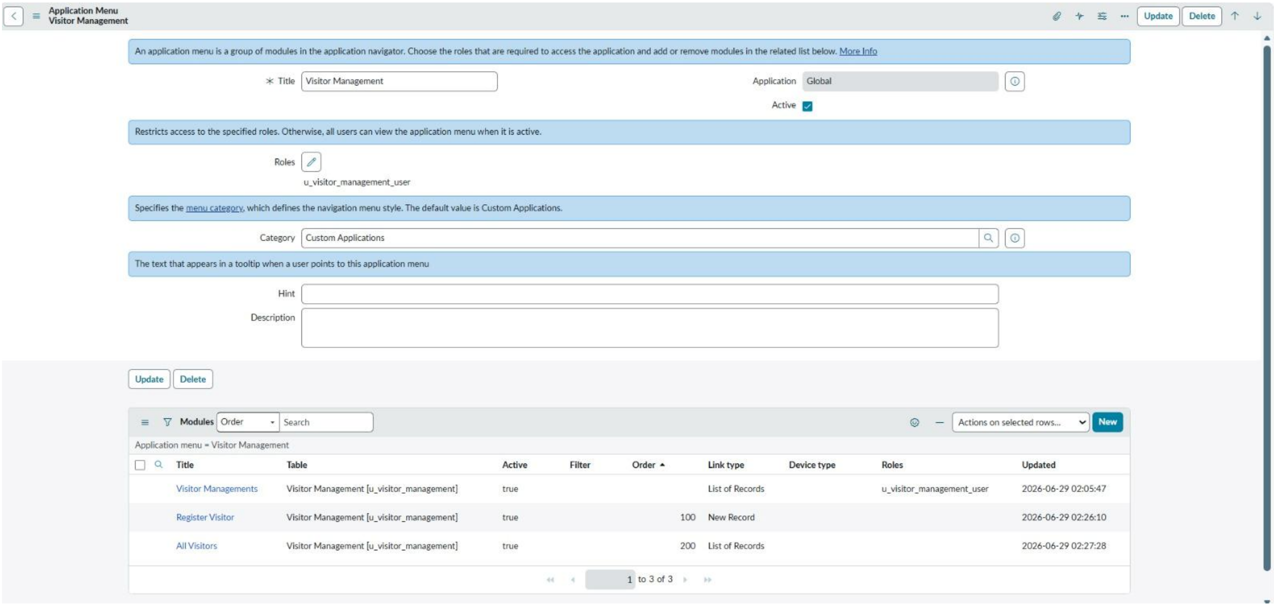
1.Log in to your ServiceNow Developer Instance.
2.Open the Application Navigator.
3.Search for Studio and open it.
4.Select the Visitor Management System application.
5.In the Application Explorer, click Create Application File.
6.Select Application Menu and enter the application details.

7.Create the following modules under the application menu:
8.Click Submit or Save to create the application menu and modules.
MODULE NAME------PURPOSE
>>Register Visitor------Used to register a new visitor.
>>All Visitors-------Displays the list of all registered visitors.









6.4 Client Script Creation (Phone Number Validation):

Purpose:

A Client Script is created to validate the visitor's phone number before the form is submitted. This validation ensures that only valid 10-digit Indian mobile numbers are accepted, preventing incorrect or incomplete data from being stored in the Visitor Management table.

Navigation

Application Navigator → System UI → Client Scripts → New

Steps to Create the Client Script

1.Log in to your ServiceNow Developer Instance.

2.Open the Application Navigator.

3.Search for Client Scripts.

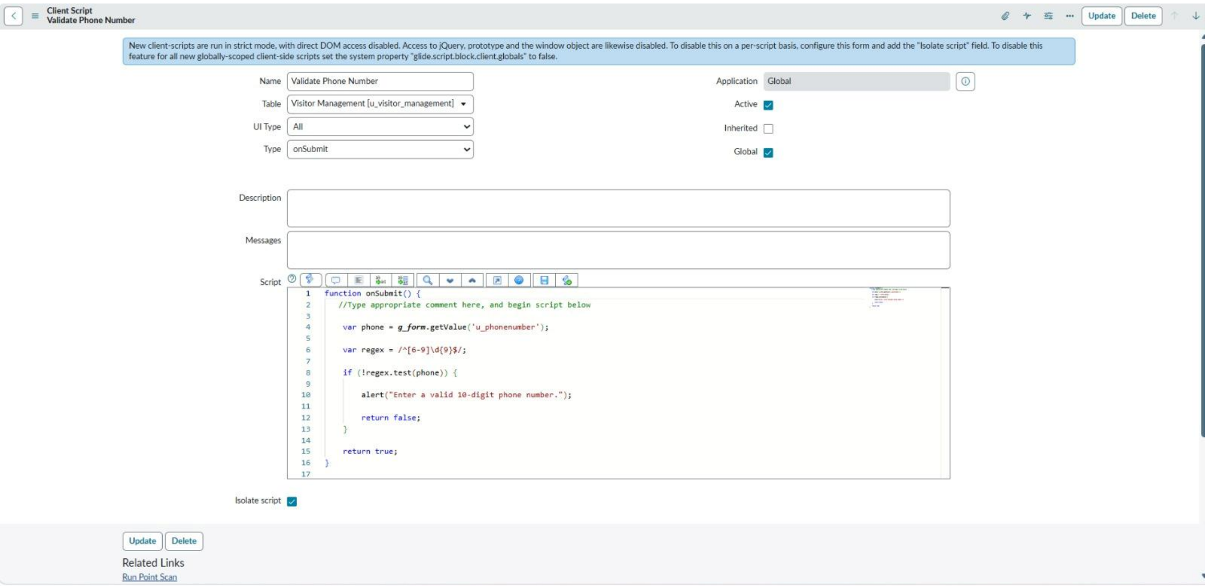
4.Navigate to System UI → Client Scripts.
5.Click the New button.
6.Enter the following details:

Field---------Value

Table--------Visitor Management
Type----------onSubmit
Name---------Validate Phone Number
Active----------Yes
In the Script section

```
function onSubmit() {
    var phone = g_form.getValue('phone_number');
    var regex = /^[6-9]\d{9}$/;
    if (!regex.test(phone)) {
        alert("Enter a valid 10-digit phone number.");
        return false;
    }
    return true;
}
```

8. Click Submit or Save to save the Client Script.



Working of the Script:

The script executes when the user clicks Submit.
It retrieves the value entered in the Phone Number field.
The regular expression checks whether the phone number starts with 6, 7, 8, or 9 and contains exactly 10 digits.

If the number is invalid, an alert message is displayed and the form submission is prevented.
If the number is valid, the record is submitted successfully.

6.5 Email Notification Creation:

Purpose:

The Email Notification feature is used to automatically notify the Security team whenever a new visitor is registered. This ensures that the security personnel are informed in advance about the visitor's arrival and can make the necessary arrangements for a smooth check-in process.

Navigation:

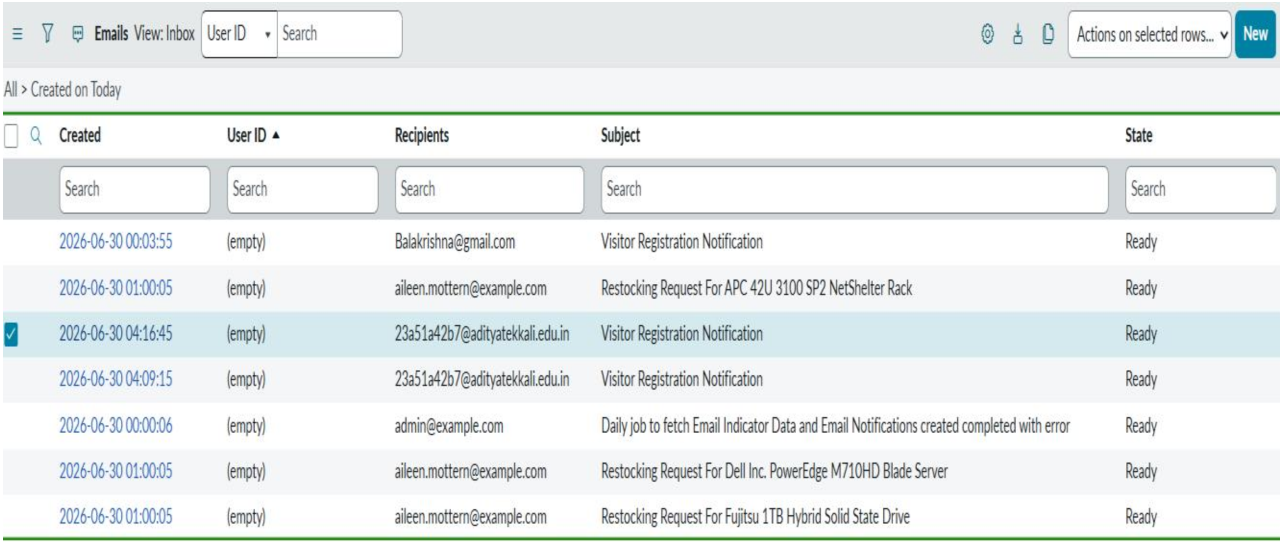
Application Navigator → System Notification → Email → Notifications → New

Steps to Create the Email Notification

1.Log in to your ServiceNow Developer Instance.
2.Open the Application Navigator.
3.Search for Notifications.
4.Navigate to System Notification → Email → Notifications.
5.Click the New button.

Field---------Value

Name------Visitor Registration
Table------Visitor Management
When to Send-----Record Inserted
Recipients------Security Group (or Specific Email Address)
Subject--------New Visitor Registered
A new visitor has been registered.
Visitor Name: ${visitor_name}
Company: ${company}
Visit Date: ${visit_date}
Host Employee: ${host_employee}



Created	User ID ▲	Recipients	Subject	State
2026-06-30 00:03:55	(empty)	Balakrishna@gmail.com	Visitor Registration Notification	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-30 04:16:45	(empty)	23a51a42b7@adityatekkali.edu.in	Visitor Registration Notification	Ready
2026-06-30 04:09:15	(empty)	23a51a42b7@adityatekkali.edu.in	Visitor Registration Notification	Ready
2026-06-30 00:00:06	(empty)	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready

Working of the Notification:

Whenever a new visitor registration record is created, the notification is triggered automatically.

The system sends an email to the Security team containing the visitor's details. This helps the Security team prepare for the visitor's arrival and improves the visitor check-in process.

6.6 UI Action Creation (Check In Button)

Purpose:

The UI Action is created to provide a Check In button on the Visitor Management form. When the button is clicked, the visitor's status is automatically updated from Registered to Checked In. This simplifies the visitor check-in process and allows security personnel or reception staff to record the visitor's arrival with a single click.

Navigation:

Application Navigator → System Definition → UI Actions → New

Steps to Create the UI Action

1.Log in to your ServiceNow Developer Instance.

2.Open the Application Navigator.

3.Search for UI Actions.

4.Navigate to System Definition → UI Actions.

5.Click the New button.

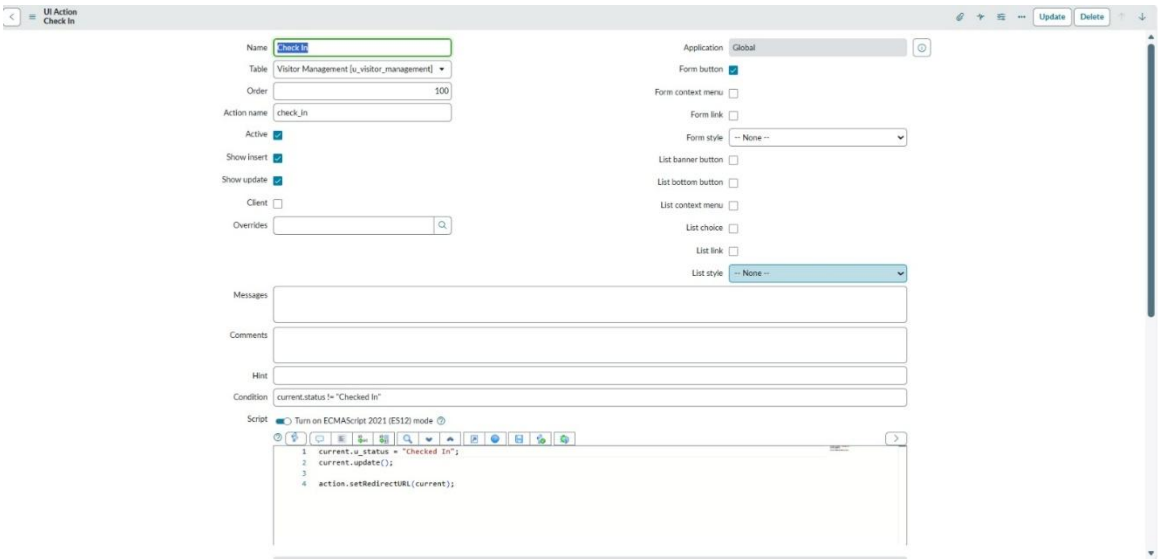
6.Enter the following details:

Field---------Value

Name---------Check In

Table----------Visitor Management

Action Name--------check_in
Form Button---------✓ Enabled
Client-----------No
Condition----------current.status != "Checked In"
In the Script field
current.status = "Checked In";
current.update();
action.setRedirectURL(current);
8. Click Submit to save the UI Action.



Working of the UI Action:

The Check In button is displayed only when the visitor's status is not "Checked In".

When the button is clicked, the system updates the visitor's status to Checked In.

The record is saved automatically, and the form is refreshed to display the updated status.

6.7 Testing and Results:

Purpose:

Testing is performed to verify that all the functionalities of the Visitor Management System work correctly. The system is tested by registering a visitor, validating the phone number, sending an email notification to the Security team, and checking in the visitor using the Check In button.

Navigation:

Application Navigator → Visitor Management System → Register Visitor

Testing Procedure

1.Open the Register Visitor module.

2.Enter the visitor details:

- Visitor Name
- Phone Number
- Email
- Company
- Visit Date
- Host Employee

3.Click the Submit button.

4.Verify that the visitor record is created successfully.

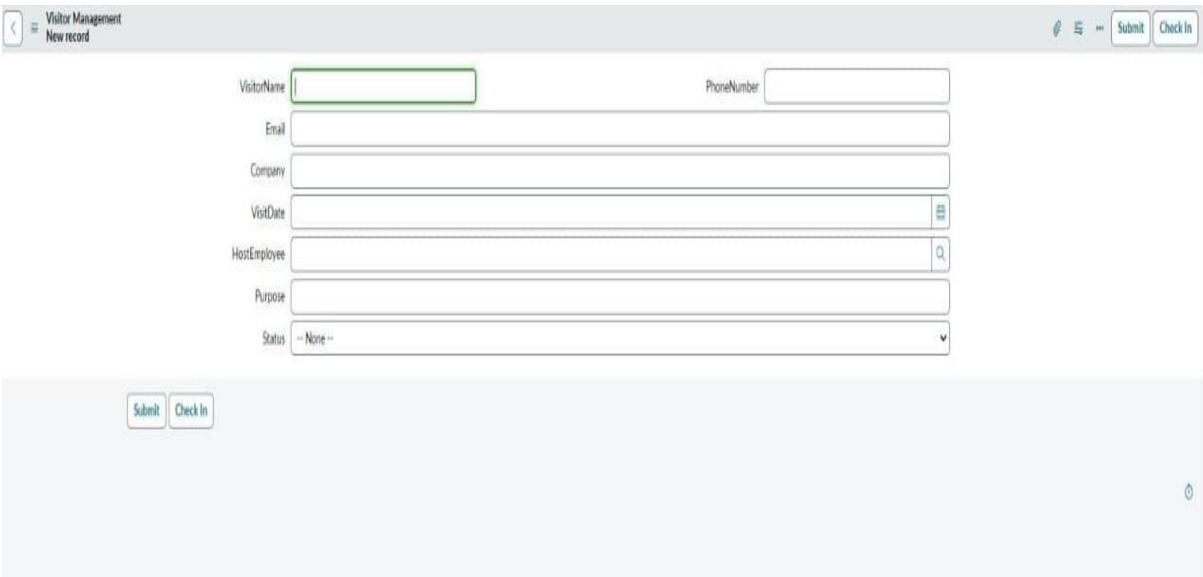
5.Enter an invalid phone number and verify that the validation message is displayed.

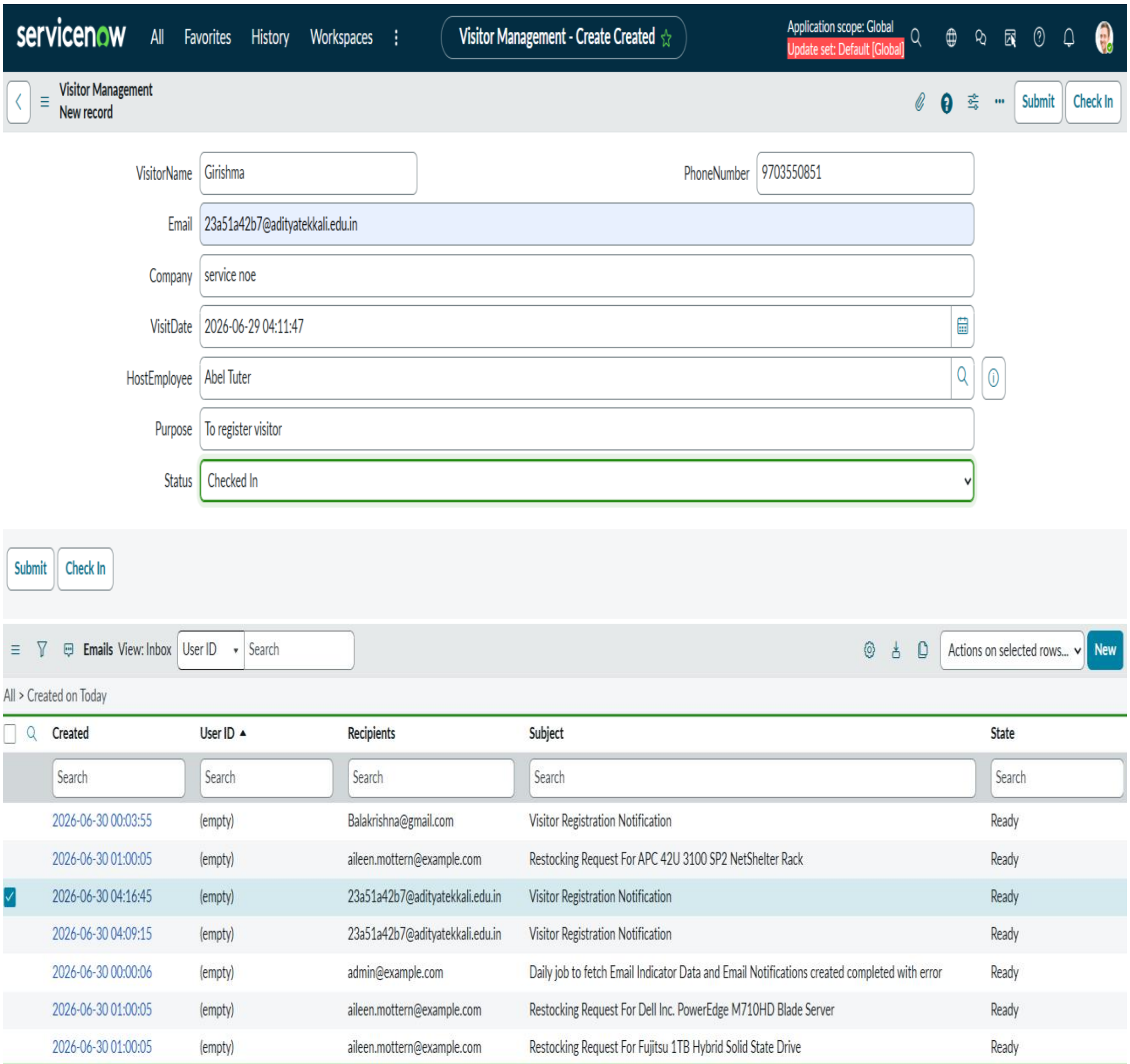
6.Check that an email notification is sent to the Security team after successful registration.

7.Open the created visitor record.

8.Click the Check In button

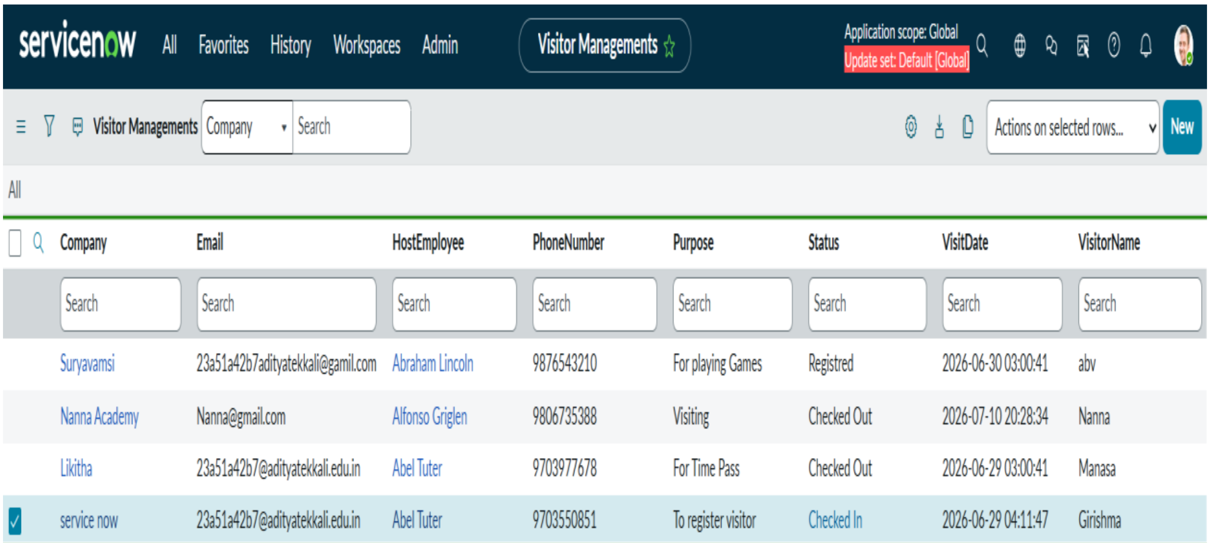
9.Verify that the Status changes from Registered to Checked In.





VisitorName	Girishma
PhoneNumber	9703550851
Email	23a51a42b7@adityatekkali.edu.in
Company	service noe
VisitDate	2026-06-29 04:11:47
HostEmployee	Abel Tuter
Purpose	To register visitor
Status	Checked In

Created	User ID	Recipients	Subject	State
2026-06-30 00:03:55	(empty)	Balakrishna@gmail.com	Visitor Registration Notification	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-30 04:16:45	(empty)	23a51a42b7@adityatekkali.edu.in	Visitor Registration Notification	Ready
2026-06-30 04:09:15	(empty)	23a51a42b7@adityatekkali.edu.in	Visitor Registration Notification	Ready
2026-06-30 00:00:06	(empty)	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-30 01:00:05	(empty)	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready



Company	Email	HostEmployee	PhoneNumber	Purpose	Status	VisitDate	VisitorName
Suryavamsi	23a51a42b7adityatekkali@gamil.com	Abraham Lincoln	9876543210	For playing Games	Registred	2026-06-30 03:00:41	abv
Nanna Academy	Nanna@gmail.com	Alfonso Griglen	9806735388	Visiting	Checked Out	2026-07-10 20:28:34	Nanna
Likitha	23a51a42b7@adityatekkali.edu.in	Abel Tuter	9703977678	For Time Pass	Checked Out	2026-06-29 03:00:41	Manasa
service now	23a51a42b7@adityatekkali.edu.in	Abel Tuter	9703550851	To register visitor	Checked In	2026-06-29 04:11:47	Girishma

Conclusion:

--The Visitor Management System was successfully designed and implemented using the ServiceNow platform. The application automates the visitor registration process, making it faster, more secure, and more efficient than traditional manual methods. Employees can easily register visitors by entering their details through a user-friendly form, and the system securely stores all visitor information in a custom table.

--The implemented features, including phone number validation using a Client Script, automatic email notifications to the security team, and the Check In UI Action, ensure data accuracy, timely communication, and smooth visitor management. The application also maintains a centralized record of all visitors, making it easy to track visitor history and monitor their status.

--This project demonstrates the effective use of ServiceNow components such as custom tables, fields, application menus, modules, client scripts, email notifications, and UI actions to develop a real-world business application. The system reduces manual effort, improves security, minimizes errors, and enhances the overall visitor management process.

--In conclusion, the Visitor Management System meets all the project requirements and provides a reliable, scalable, and efficient solution for managing visitors in an organization. It can be further enhanced by adding features such as visitor check-out, QR code generation, ID card scanning, approval workflows, SMS notifications, and visitor reports to make the system more comprehensive and user-friendly.